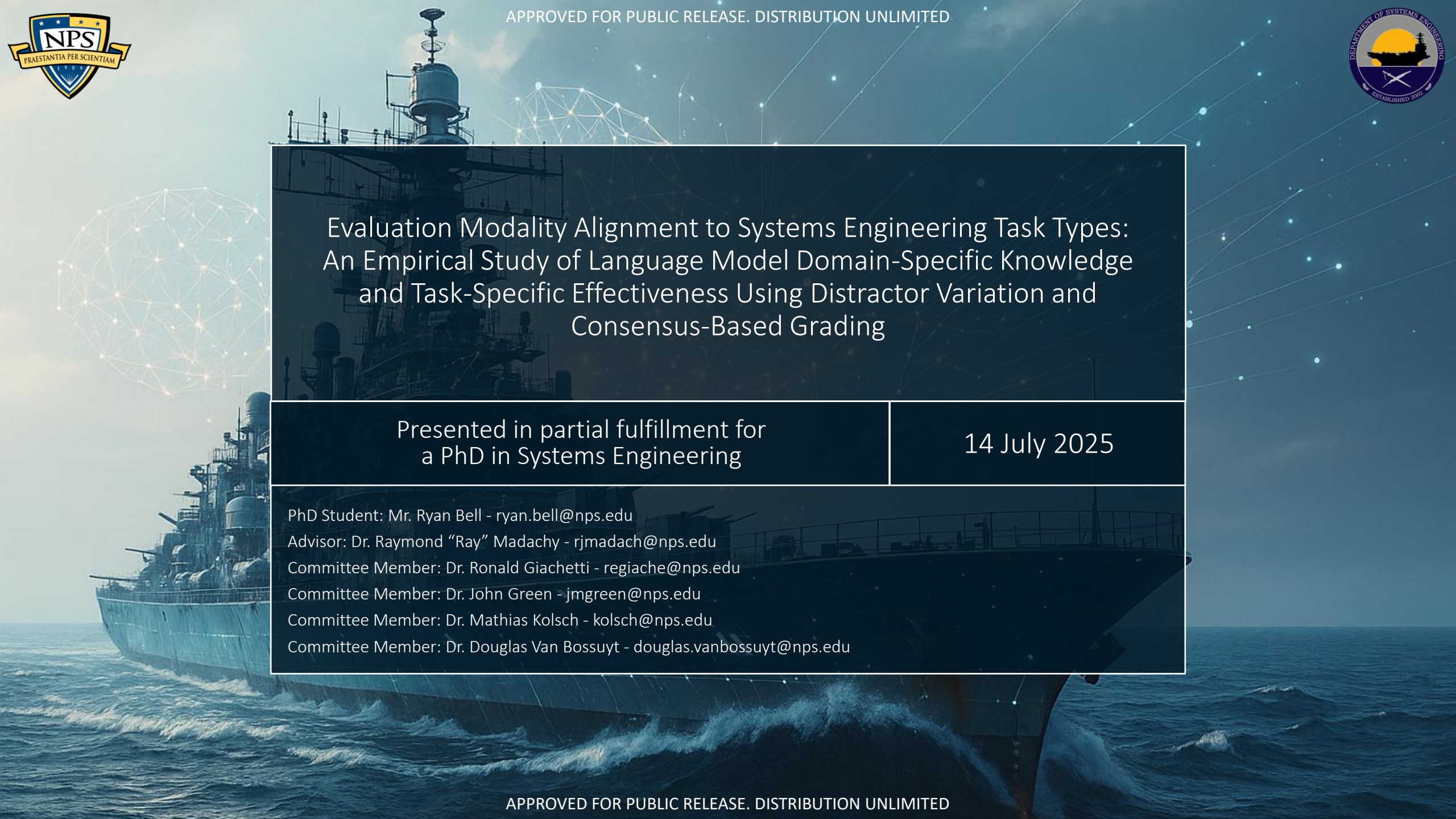




Evaluation Modality Alignment to Systems Engineering Task Types: An Empirical Study of Language Model Domain-Specific Knowledge and Task-Specific Effectiveness Using Distractor Variation and Consensus-Based Grading

Presented in partial fulfillment for
a PhD in Systems Engineering

14 July 2025

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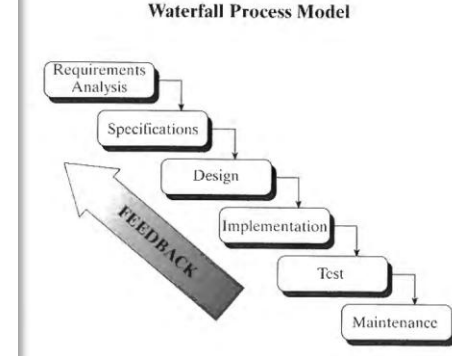
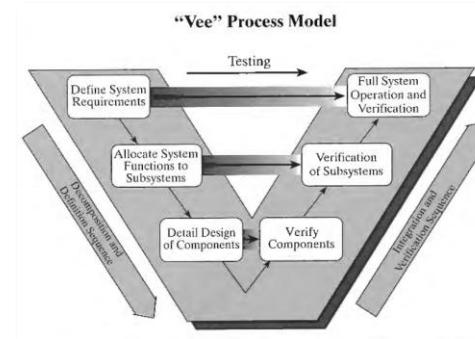


Agenda

1. Background
 - Systems Engineering
 - AI and Language Models
 - Benchmarking
2. Introduction
 - Problem Statement
 - Research Questions
 - Objective
3. Research Approach
 - Literature Review
 - Research Framework
 - Analysis Methods
4. Research Contributions
5. Execution Plan
 - Dissertation Outline
 - Dissertation Schedule
 - Notional Publications and Conference Paper
6. Committee Management Plan

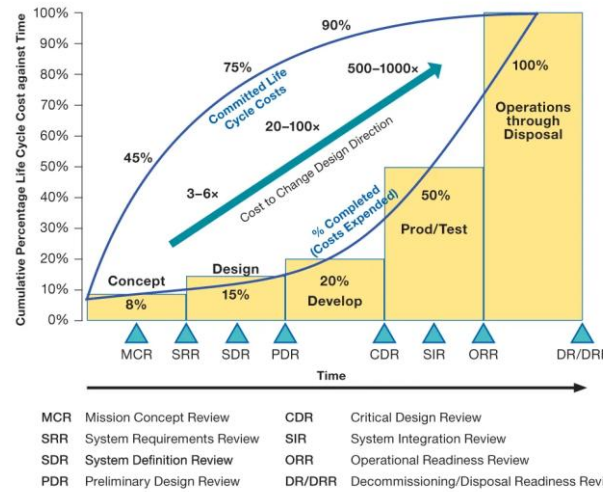
Background: Systems Engineering

- Systems engineers typically follow **process models**, including:
 - “Vee” Model
 - Waterfall Model
 - Spiral Model
- Goal is to balance **performance**, **schedule**, and **cost** of the effort
- We classify and bucket Systems Engineering processes in IEEE 15288



Courtesy of: Systems Engineering and Analysis Blanchard and Fabrycky

Engine



Adapted from INCOSE-TP-2003-002-04, 2015

Technical processes		Technical management processes	Agreement processes	Organizational project-enabling processes
Business or mission analysis process	Integration process	Project planning process	Acquisition process	Life cycle model management process
Stakeholder needs & requirements definition process	Verification process	Project assessment and control process	Supply process	Infrastructure management process
System requirements definition process	Transition process	Decision management process		Portfolio management process
Architecture definition process	Validation process	Risk management process		Human resource management process
Design definition process	Operation process	Configuration management process		Quality management process
System analysis process	Maintenance process	Information management process		Knowledge management process
Implementation process	Disposal process	Measurement process		
		Quality assurance process		

FIGURE 1.1 System life cycle processes per ISO/IEC/IEEE 15288. This figure is excerpted from ISO/IEC/IEEE 15288:2015, Figure 4 on page 17, with permission from the ANSI on behalf of the ISO. © ISO 2015. All rights reserved.

Courtesy of: *Systems Engineering Handbook: A Guide For Systems Life Cycle Processes and Activities*, 4th Edition, Wiley

Background: AI and Language Models

- **Large Language Models (LLMs)** are neural networks that have revolutionized the field of natural language processing (NLP) by demonstrating an impressive ability to understand and generate text. Many models have also demonstrated an ability to generate code in common languages (e.g. Python).
- **Training:** Teaching a model how to perform a task using large amounts of data
- **Inferencing:** Model making inferences based on new inputs
- **Common LLMs** (not an exhaustive list)
 - Proprietary: OpenAI ChatGPT, Google Gemini, Anthropic Claude
 - Open source: Llama, Falcon, Mistral
- **Model Characteristics**
 - Size is the number of parameters (e.g. 70B parameters)
 - Quantization is a compression technique used to reduce memory and compute requirements. It converts weights to a lower precision data type (e.g. 32 bit floating point → 8 or 4 bit integer).
- **Challenges**
 - Hallucinations
 - Access to unique or domain-specific training data
 - Inelegant or not optimized code generation



Artificial Intelligence

Is the field of study

Machine Learning

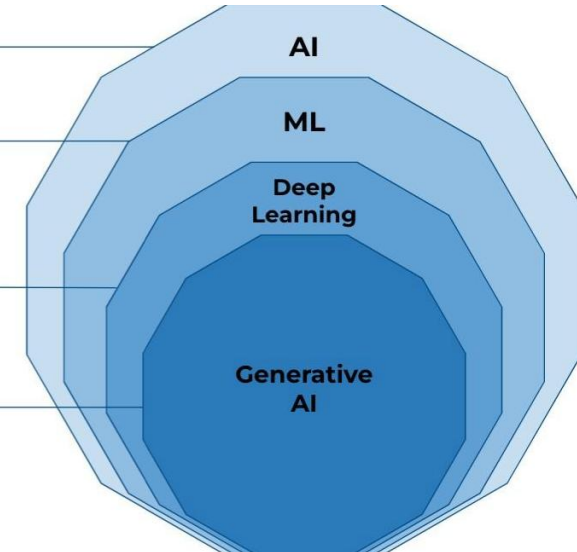
Is a branch of AI that focus on the creation of intelligent machines that learn from data. Another very well know branch inside AI is **Optimization**.

Deep Learning

Is a subset of Machine Learning methods, based on **Artificial Neural Networks**.
Examples: CNNs, RNNs

Generative AI

A type of ANNs that generate data that is similar to the data it was trained on.
Examples: GANs, LLMs





Background: AI and Language Models, Improving Responses

- 
- **Re-Training:** Re-train base model with new dataset
 - **Finetuning:** Teach the base model specialized knowledge by adjusting a small number of internal parameters. Significant time savings compared to re-training entire model
 - **Retrieval Augmented Generation (RAG):** Utilize external data to augment the LLM's knowledge without changing model weights
 - **Prompt Engineering:** Optimize prompts for optimal outputs
 - **Few Shot** prompts teach LLMs to perform a task through examples (Oleszak, 2024)
 - **Zero Shot** prompts challenge the LLM to perform a task correctly when it has not been correctly trained for that task

These methods can also be combined.

Background: Benchmarking Language Models

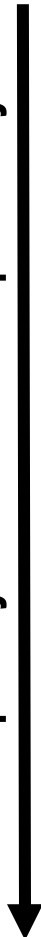
■ Benchmarks have evolved over time

- Early benchmarks focused on foundational tasks such as **work relationships** and their **semantic similarities**
- More recently, more complex benchmarks have been developed such as College Medicine, Physics, Biology, Comp Sci, Math, Electrical Engineering, among others

■ Complexity and domain specificity of the benchmarks over time have been indicative of progression

- Almost all listed here are focused on **natural language processing**, few vision model questions

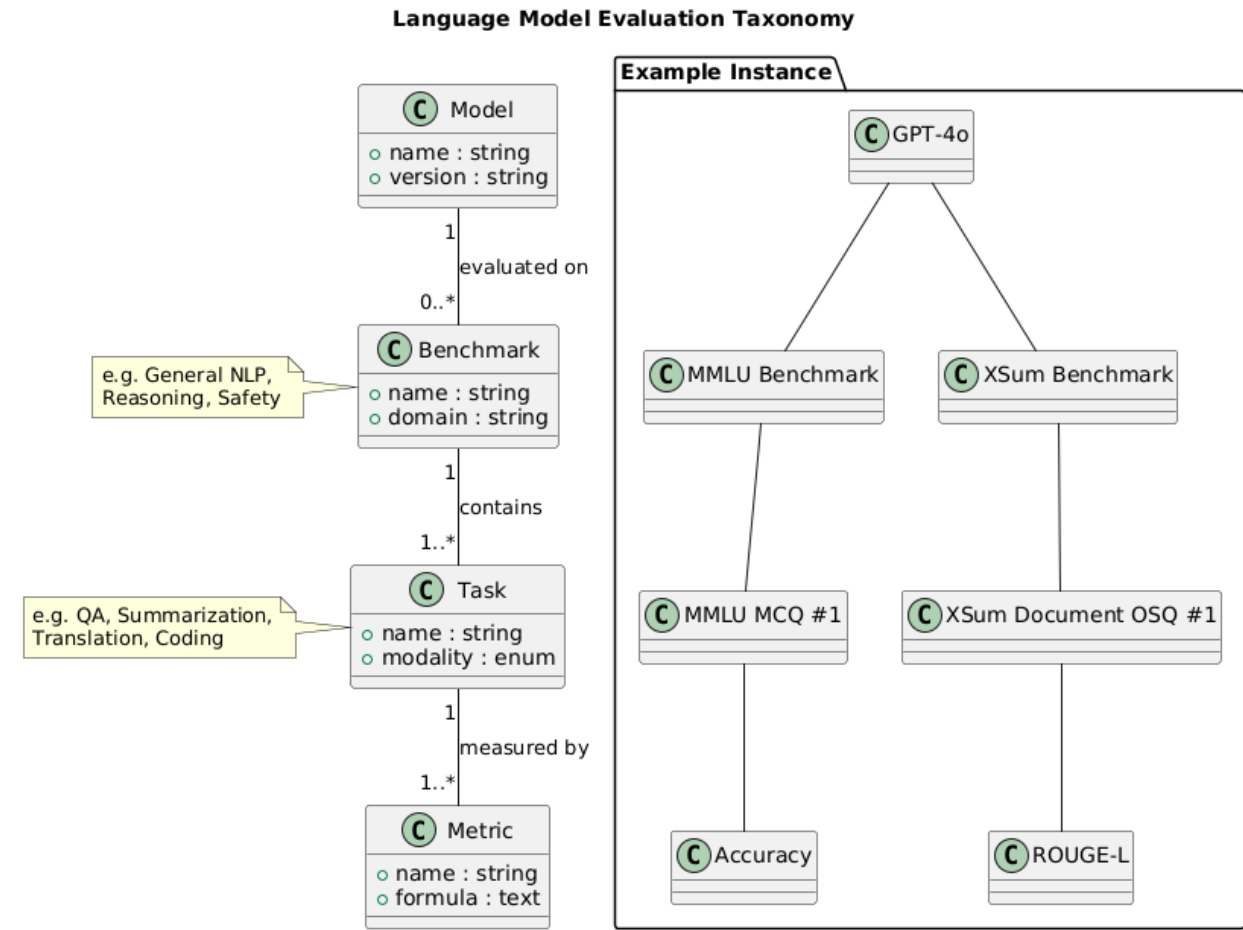
Complexity and Domain Specificity



Benchmark Name	Topic	Released
WordNet	Word relationships and meanings, foundational dataset for semantic similarity and language understanding	1985
MNIST	Handwritten digit recognition, foundational for image processing and computer vision	1998
BLEU	Language translation quality metric, foundational for evaluating machine translation systems	2002
Enron Emails	Recognizing names, entities, and information extraction from natural email datasets	2004
ImageNet	Large-scale image recognition and classification, pivotal in advancing deep learning in computer vision	2009
LAMBADA	Understanding context and reasoning in text, focusing on predicting sentence endings	2016
SWAG	Common sense reasoning and predicting plausible sentence endings in a given context	2018
GLUE	A collection of diverse NLU tasks like question answering and sentiment analysis to advance language understanding across various contexts.	2018
SuperGLUE	A successor to GLUE with more challenging tasks, pushing the limits of NLU models with advanced reasoning and co-reference resolution.	2019
HellaSWAG	An extension of SWAG for more challenging common sense reasoning scenarios	2019
ARC	ARC evaluates an AI's ability to tackle each task from scratch, using only the kind of prior knowledge about the world that humans naturally possess, known as core knowledge.	2019
DROP	Reasoning over paragraphs, requires numerical reasoning and understanding of natural language	2019
Winogrande	A large-scale dataset of winograd schemas designed to improve commonsense reasoning in AI systems.	2019
XTREME	Cross-lingual understanding and translation across multiple languages, tests multilingual capabilities	2020
MMLU	Measures professional and academic knowledge across various fields including College Medicine, Physics, Biology, Comp Sci, Math, Electrical Engineering, Professional Accounting, Psychology and worldly knowledge about Foreign Policy and Religions, among others	2021
TruthfulQA	A question-answering dataset designed to evaluate a model's ability to produce truthful and factual answers.	2021
GSM8K	Grade School Math 8K (GSM8K), a collection of math word problems aimed at evaluating numerical reasoning.	2021
BIG-Bench	Broad spectrum of tasks testing reasoning, common sense, professional knowledge, and language capabilities	2022

Background: Benchmarking versus Metrics

- **Benchmark:**
 - A *set of inputs* (prompts, questions, documents) with *ground-truth outputs* and a protocol that defines how the model should interact with them.
 - Examples:
 - Multitask Multiple Choice (**MMLU**)
 - Grade-School Math (**GSM8K**)
- **Metric:**
 - A *formula* that turns the model's outputs on a task into a *number* (score).
 - Examples:
 - Bilingual Evaluation Understudy (**BLEU**)
 - Recall-Oriented Understudy for Gisting Evaluation (**ROUGE**)
- **Takeaway:**
 - A *task* is **what** the model is asked to do; a *metric* is **how** we measure whether it did it well.



Background: The Benchmarking Process

- There are several different frameworks and tools for model evaluation.
- The selection of which framework depends on what you are trying to achieve.

Aspect	LM-Eval-Harness	OpenAI Evals	Hugging Face Evaluate	Chatbot Arena
Primary Usage	Generic NLP tasks, accuracy	OpenAI-specific evals	Metrics computation	Dialogue/chatbot comparisons
Benchmark Tasks	60+ standardized tasks	Official + custom user evals	None (metrics-focused only)	Conversational pairs
Human Judgement	Minimal/None	Optional	Minimal/None	Core component
Metrics Provided	Accuracy, EM scores	Custom-defined (scores/pass-fail)	Rich metrics (BLEU, ROUGE, F1)	Win rates, Elo ratings
Setup complexity	Medium/High	Low (managed)	Low (metric-only)	Low (web-based)
Subjectivity	Low (metric-driven)	Low/High (depends on test)	Low (metric-driven)	High (human judges)
Integration	Hugging Face, APIs, CLI	OpenAI ecosystem	Hugging Face Hub	Web-based UI
Reproducibility	High	High	High	Low (human variability)
Customizability	High	High	High	Medium (custom pairs)
Best For	Research, standard NLP tasks	Quick OpenAI API checks	Detailed metric analysis	Real-world chatbot quality



Background: Example Community Leaderboards

- Results of benchmarks are typically published in papers and uploaded to leaderboards
- HuggingFace leaderboards for open source language models and the Chatbot Arena

Open LLM Leaderboard

Hugging Face

Search models, datasets, users...

Models Datasets Spaces Community Docs Enterprise Pricing

Open LLM Leaderboard Enterprise Community

https://huggingface.co/spaces/open-llm-leaderboard

Activity Feed Follow 1,066

AI & ML interests

Evaluating open LLMs

Recent Activity

alozowski new activity 8 days ago
open-llm-leaderboard/open_llm_leaderboard: please...

thomwolf authored a paper 8 days ago
SmolVLA: A Vision-Language-Action Model for Affordable and ...

clefourrier new activity 27 days ago
open-llm-leaderboard/results: Create Emma_R1

View all activity

Team members 18

Organization Card

Open LLM Leaderboard

This is the hub organisation maintaining the [Open LLM Leaderboard](#).

In this space you will find the dataset with detailed results and queries for the models on the leaderboard.

Score results are [here](#), and current state of requests is [here](#). For the detailed prediction, look for your model name in the datasets below!

Collections 3

Details

A gated collection of datasets containing evaluation details

open-llm-leaderboard/tensopolis__virtuoso-lite-tensop...
Viewer · Updated Mar 9 · 43.2k · 33

open-llm-leaderboard/tensopolis__falcon3-10b-tensop...
Viewer · Updated Mar 8 · 43.2k · 28

open-llm-leaderboard/Pinkstack__SuperThoughts-CoT-14...
Viewer · Updated Feb 13 · 43.2k · 8 · 2

open-llm-leaderboard/prithivMLmods__QwQ-LCoT-14B-Con...

Open LLM Leaderboard 2

Open LLM Leaderboard 🏆
Track, rank and evaluate open LLMs and chatbots

Open-LLM performances are plateauing, let's make the leaderboard steep again
Update leaderboard for fair model evaluation

open-llm-leaderboard/contents
Viewer · Updated Mar 20 · 4.58k · 9.74k · 16

open-llm-leaderboard/results

Expand 3 collections

Spaces 4

Sort: Recently updated

Open LLM Leaderboard

Track, rank and evaluate open LLMs and chatbots

Open LLM Leaderboard Model Comparator

Compare Open LLM Leaderboard results

Chatbot Arena LLM Leaderboard

Spaces Imarena.ai / chatbot-arena-leaderboard like 4.46k Running

Chatbot Arena LLM Leaderboard: Community-driven Evaluation for Best LLM and AI chatbots

Twitter | Discord | Blog | GitHub | Paper | Dataset | Kaggle Competition

This is a mirror of the live leaderboard created and maintained at <https://lmarena.ai/leaderboard>. Please link to the original URL for citation purposes.

Chatbot Arena (lmarena.ai) is an open-source platform for evaluating AI through human preference, developed by researchers at [LMarena](https://lmarena.ai). With over 1,000,000 user votes, the platform ranks best LLM and AI chatbots using the Bradley-Terry model to generate live leaderboards. For technical details, check out our [paper](#).

Chatbot Arena thrives on community engagement — cast your vote to help improve AI evaluation!

New Arena UI at lmarena.ai/leaderboard! Check it out and give feedback!

Arena NEW: Overview Arena (Vision) Arena-Hard-Auto Full Leaderboard

Total #models: 242. Total #votes: 3,021,304. Last updated: 2025-06-04.

Code to recreate leaderboard tables and plots in this [notebook](#). You can contribute your vote at lmarena.ai!

Category Overall

Apply filter

Style Control Show Deprecated

Overall Questions

#models: 242 (100%) #votes: 3,021,304 (100%)

Rank* (UB)	Rank (StyleCtrl)	Model	Arena Score	95% CI	Votes	Organization	License	Knowledge Cutoff
1	1	Gemini-2.5-Pro-Preview-06-05	1476	+8/-8	4701	Google	Proprietary	Unknown
2	2	Gemini-2.5-Pro-Preview-05-06	1446	+5/-7	10386	Google	Proprietary	Unknown
3	5	Gemini-2.5-Flash-Preview-05-20	1420	+5/-9	9970	Google	Proprietary	Unknown
3	2	o3-2025-04-16	1420	+4/-4	13888	OpenAI	Proprietary	Unknown
3	4	ChatGPT-4o-latest (2025-03-26)	1417	+4/-4	10302	OpenAI	Proprietary	Unknown
4	8	Grok-3-Preview-02-24	1412	+4/-4	20585	xAI	Proprietary	Unknown
6	4	GPT-4.5-Preview	1406	+5/-4	15271	OpenAI	Proprietary	Unknown
8	8	Gemini-2.5-Flash-Preview-04-17	1392	+4/-5	13320	Google	Proprietary	Unknown
9	8	GPT-4.1-2025-04-14	1377	+5/-5	12421	OpenAI	Proprietary	Unknown
9	10	DeepSeek-V3-0324	1377	+4/-4	15122	DeepSeek	MIT	Unknown
9	16	Hunyuan-Turbos-20250416	1366	+9/-8	5230	Tencent	Proprietary	Unknown
10	6	Claude-Opus_4 (20250514)	1366	+7/-5	10402	Anthropic	Proprietary	Unknown
11	11	DeepSeek-R1	1366	+4/-4	19430	DeepSeek	MIT	Unknown



Background: AI Performance Gains

A brief survey on AI related performance gains yields:

- Gains from AI are significant across domains, ranging from a mere 10% (typically more complex tasks) to claiming 10x (on monotonous tasking).
- Systems Engineering productivity gains have not been assessed, yet...

Engineering domain	Representative AI tool / Use case	Reported Productivity or Performance Gain	Source
Software engineering	GitHub Copilot pair-programming	55 % faster completing a JS server task in controlled experiment	https://arxiv.org/abs/2302.06590
	Generative-AI code assistants (multi-company study)	Up to 2 × faster on coding tasks	https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/unleashing-developer-productivity-with-generative-ai
	Google internal AI agents	10 % higher engineering throughput; AI writes 30 % of new code	https://research.google/blog/ai-in-software-engineering-at-google-progress-and-the-path-ahead/#footnote-item-2
Civil / site design	Bentley OpenSite+ generative layout & grading	Up to 10 × faster producing viable site plans	https://www.bentley.com/news/bentley-systems-announces-generative-ai-game-changer-for-civil-site-design
Mechanical / product CAD	ML- and generative-design-driven CAD workflows	Significantly reduces design time (qualitative review across multiple case studies)	https://www.researchgate.net/publication/390640817_ARTIFICIAL_INTELLIGENCE_IN_THE_CAD_PROCESS_MACHINE_LEARNING_MODELS_GENERATIVE_OPTIMISATION_AND_THEIR_IMPACT_ON_DESIGN
Manufacturing / quality	AI vision inspection for defect detection	Up to 50 % productivity increase on inspection lines	https://landing.ai/wp-content/uploads/2020/11/MachineVisionSurvey.pdf
Enterprise software services	Cognizant AI coding rollout	34 – 37 % jump in bottom-quartile developer output	https://www.businessinsider.com/cognizant-ceo-entry-level-hiring-job-market-grow-ai-2025-6

Background: Systems Engineering ROI

“SE effort has a significant, quantifiable effect on program success, with correlation factors as high as 80%”

... So what if we applied AI performance gains to SE?

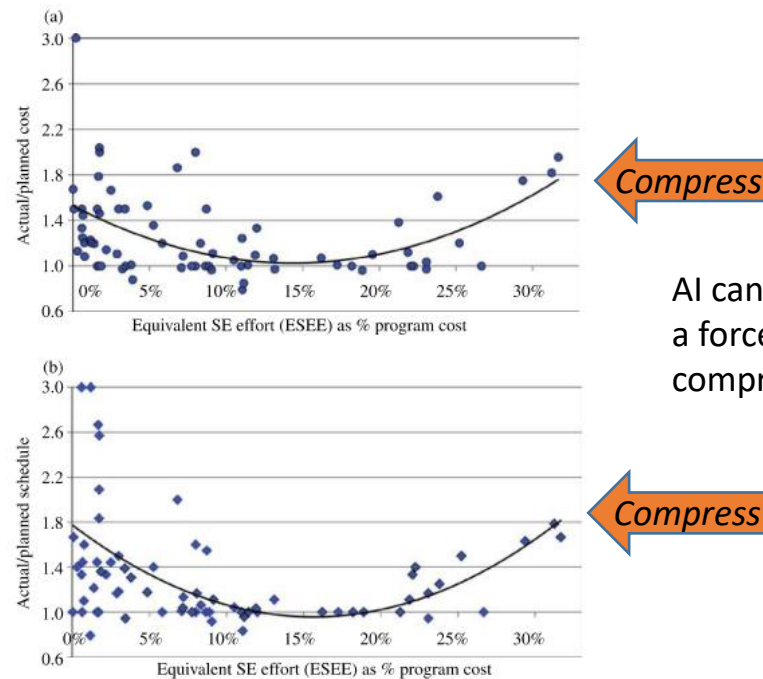


FIGURE 2.7 Cost (a) and schedule (b) overruns correlated with SE effort. From Honour (2013). Reprinted with permission from Eric Honour. All other rights reserved.

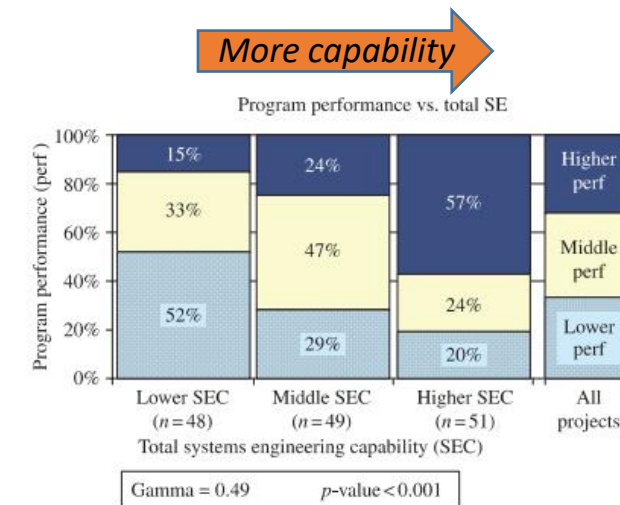


FIGURE 2.6 Project performance versus SE capability. From Elm and Goldenson (2012). Reprinted with permission from Joseph Elm. All other rights reserved.

Courtesy of: Systems Engineering Handbook: A Guide For Systems Life Cycle Processes and Activities, 4th Edition, Wiley



Hypothesis

AI can help push systems engineers and teams to greater capability, and thus higher project performance.



Problem Statement

How do we assess and compare performance of Large Language Models (LLMs) in the field of Systems Engineering?



Primary Research Questions

- How do the evaluation modalities of MCQs and textual responses compare in effectively assessing the performance of LLMs in systems engineering?
- What are the strengths and limitations of using MCQs as a benchmarking methodology in a specialized domain such as systems engineering?
- How can textual response formats better capture the nuanced reasoning and conceptual understanding required for systems engineering tasks?



Supporting Research Questions

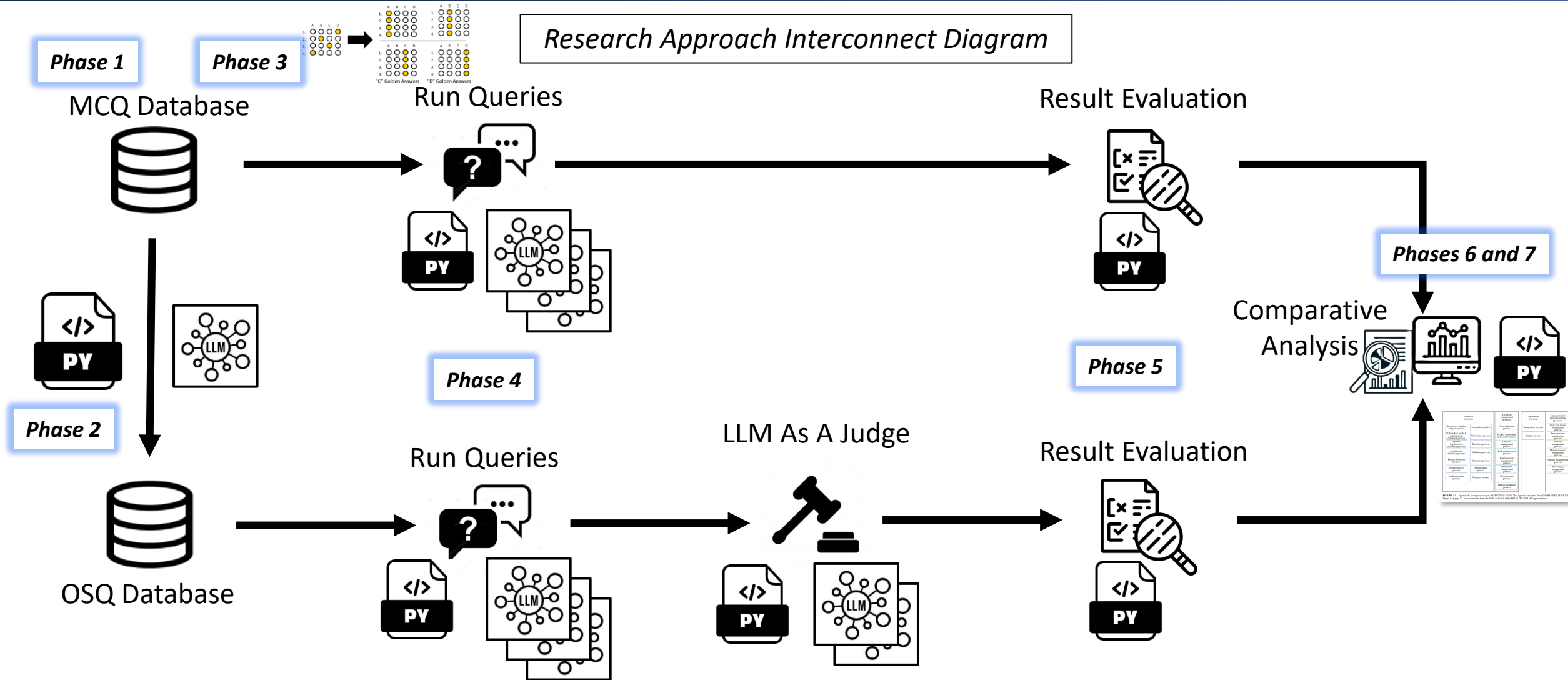
- What is the current state of benchmarking methodologies for LLMs in systems engineering?
- What is the role of textual response formats in capturing deeper conceptual understanding compared to MCQs?
- How can a systematic approach be developed to convert MCQs into high-quality textual datasets while preserving semantic fidelity?
- What evaluation metrics can effectively compare LLM-generated textual responses against reference solutions?
- How can insights gained from the comparative evaluation of MCQs and textual responses inform the design of more effective benchmarks for LLMs in systems engineering?

Research Approach: Literature Review

Section ID	Chapter Title	Description
L.1	Introduction and Context Setting	• Motivation for LLMs given productivity benefits. • Summary of recent LLM advances and growing relevance to engineering. • Overview of SE complexity and representative tasks (requirements, traceability, risk). • Need for SE-specific benchmarking.
L.2	Current Benchmarking Landscape	• General benchmarks: HELM, MMLU, ARC, BIG-bench. • Reasoning benchmarks: GSM8K, ProntoQA, multi-step tasks. • Domain-specific: SWE-bench, LegalBench, MedQA, PubMedQA, Rosetta-PL. • Gaps in addressing SE-specific workflows and evaluation. • Introduction to SysEngBench structure and INCOSE Handbook categories.
L.3	Evaluation Methods for LLMs	• MCQs: strengths, classical psychometrics, distractor analysis. • OSQs: inter-rater reliability (Cohen's/Fleiss' Kappa), LLM-as-a-Judge grading. • Hybrid and purpose-built evaluation methods.
L.4	Unique Evaluation Challenges in SE and Cross-Domain Strategies	• Logical reasoning across SE views and dependencies. • Diagrammatic and formal notation reasoning (e.g., BDDs, state machines). • Numeric reasoning challenges and safety-critical applications. • Traceability, iterative refinement, and requirement consistency. • Cross-domain inspirations from law (citation grounding), software (code verification), and medicine (hallucination handling).
L.5	Experimental Designs for Benchmark Comparison	• Methods for converting MCQs to OSQs. • Statistical comparison techniques (t-test, Wilcoxon). • Case studies and lessons learned from benchmark transformations.
L.6	Measurement and Analysis Techniques	• Classical test theory: item difficulty, discrimination indices. • Item Response Theory (IRT): capability modeling. • Validity and reliability: inter-rater agreement, grounding, and test fidelity.

Table 2.1. Proposed Literature Review Outline

Research Approach: Overview

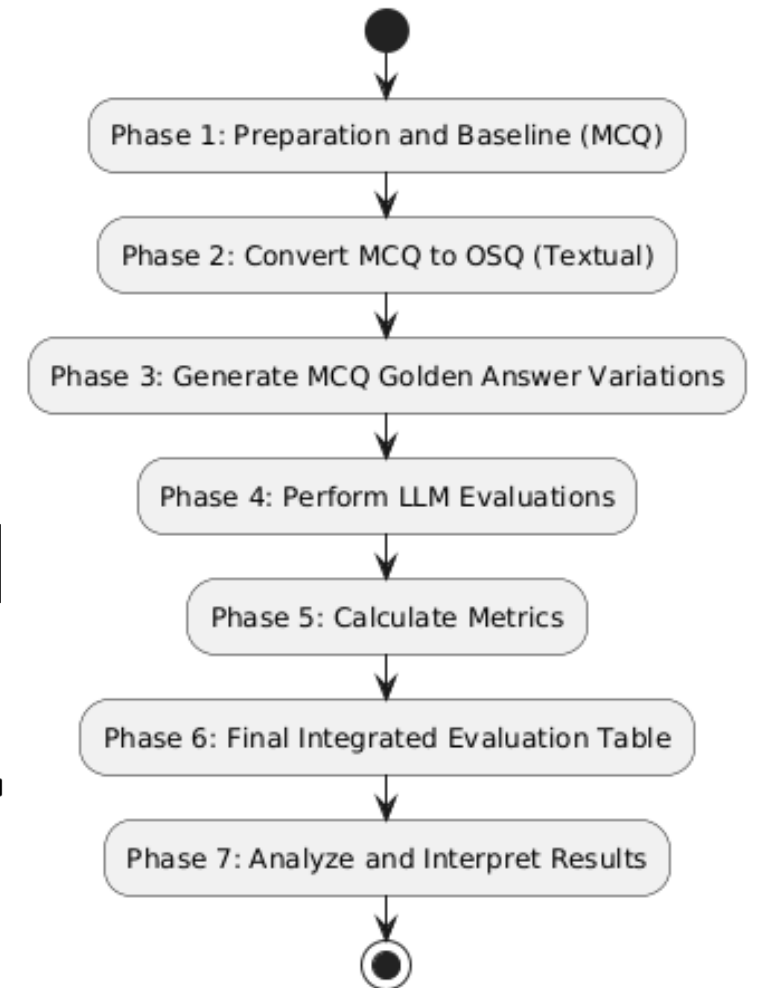


Research Approach: Overview

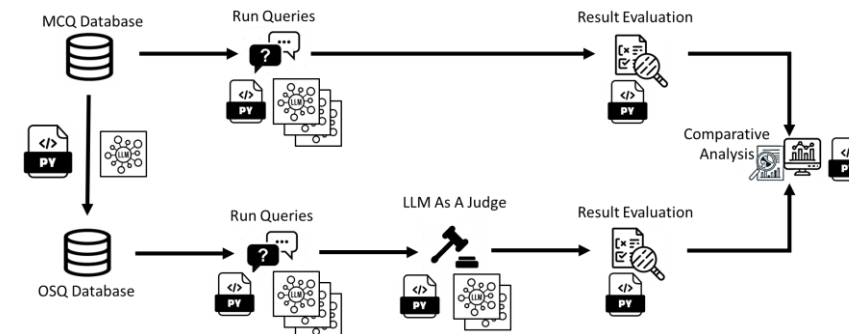
- There 7 Phases are:
 - Phase 1: Preparation and Baseline (MCQ)
 - Phase 2: Convert MCQ → OSQ (Textual)
 - Phase 3: Generate MCQ Golden Answer Variations
 - Phase 4: Perform LLM Evaluations
 - Phase 5: Calculate Metrics
 - Phase 6: Final Integrated Evaluation Table
 - Phase 7: Analyze and Interpret Results

- Let's break down each phase...

Research Approach Activity Diagram



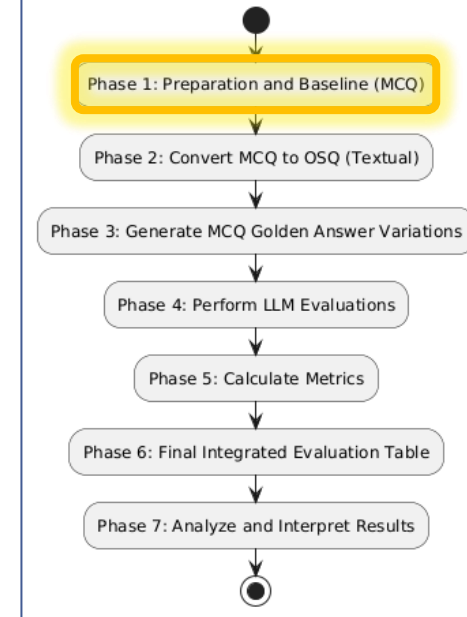
Research Approach Interconnect Diagram



Research Approach – Phase 1: Preparation and Baseline

- Down selection process for the research benchmarking framework.
- Lm-eval-harness is the selection of choice, given the criterion.

Criterion	LM-Eval-Harness	OpenAI Evals	HF Evaluate	Chatbot Arena
Structured task support	✓ MCQ, fill-in-the-blank, logic, math reasoning	✓ MCQ, fill-in-the-blank, logic, math reasoning	✗ Metrics only	✗ Conversational pairs only
Objective metrics (e.g. accuracy)	✓ Built-in, reproducible	✓ Built-in, reproducible	✓ Metric-rich, but no model control	✗ Relies on subjective human votes
Minimal human judgment	✓ Fully automatic scoring	✓ Fully automatic scoring	✓ Fully automatic scoring	✗ Human-in-the-loop is core
Reproducibility	✓ Deterministic and scriptable	✓ High with config control	✓ High (metrics are deterministic)	✗ Human variance = non-reproducible
Custom task/metric integration	✓ Full support (e.g. SysEngBench JSON/CSV)	⚠ Custom templates allowed, but tied to OpenAI infra	⚠ With HF support	⚠ Limited to conversational prompts
Offline / local model support	✓ Hugging Face, GGUF, APIs, CLI	✗ Requires OpenAI API access	✓ Works with your outputs	✗ No local model support
Best for benchmarking research	✓ Academic-grade, open-source, and modular	⚠ Fast dev tests, but not ideal for research	✓ Great for metric comparison only	✗ Not suitable for systematic benchmarking



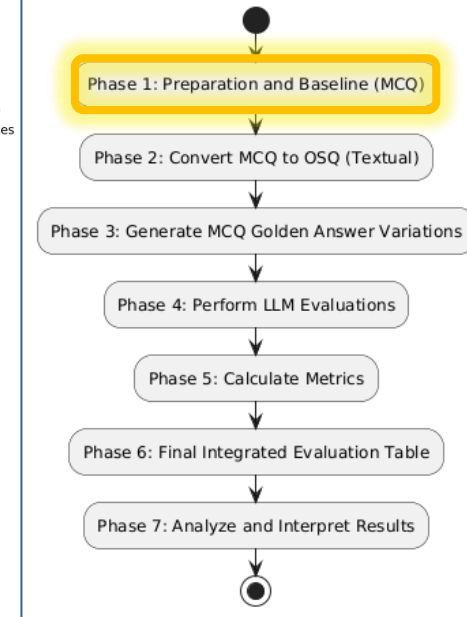
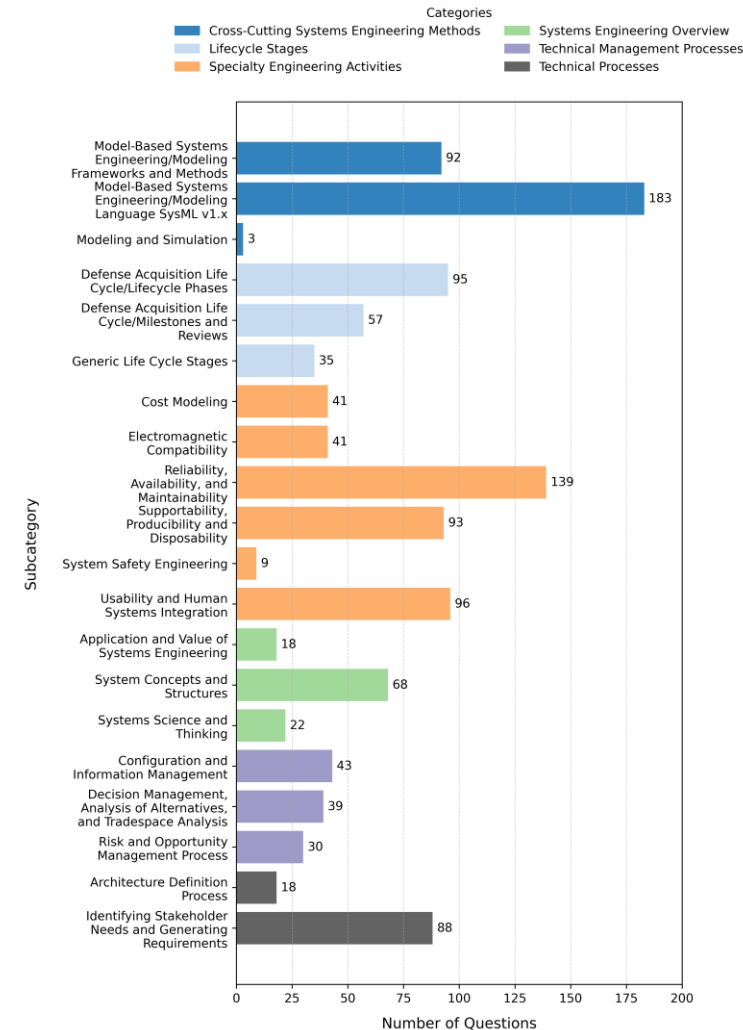


Research Approach – Phase 1: Preparation and Baseline

- Hybrid construction of by hand and synthetic data
- Reviewed by practicing Systems Engineers
- Derived from and Mapped to INCOSE HB Chapters & Topics

TABLE 1 INCOSE Handbook Chapters Incorporated into SysEngBench

Chapter	Included?	Reasoning
Chapter 1	No	This chapter provides an overview of the handbook itself, which is introductory and not directly relevant to fundamental SE concepts.
Chapter 2	Yes	Provides foundational knowledge of system concepts, structures, and systems thinking, which align with core SE principles.
Chapter 3	Yes	Discusses life cycle stages and approaches, which are essential for understanding SE project management and progression.
Chapter 4	Yes	Covers technical processes, including requirements gathering, architecture definition, and system analysis, which are fundamental to SE practices.
Chapter 5	Yes	Includes technical management processes such as decision management, risk management, and configuration management, which are key to effective SE project management.
Chapter 6	No	Focuses on the specifics of contracting and procurement processes, which are more related to acquisition than to general SE knowledge.
Chapter 7	No	Discusses typical organizational project-enabling processes that can be mapped later, as they have limited direct content related to core SE concepts.
Chapter 8	No	Details tailoring SE processes for specific sectors or domains, which is too specialized for general SE knowledge and can be considered later if needed.
Chapter 9	Yes	Focuses on cross-cutting SE methods such as modeling, simulation, and MBSE, which are integral to modern SE practices.
Chapter 10	Yes	Covers specialty engineering activities like cost modeling, reliability, safety, and human systems integration, which are critical for comprehensive SE understanding.



Research Approach – Phase 1: Preparation and Baseline

- SysEngBench created, 35 LLMs benchmarked and placed on the Pareto curve
- Efficiency spotlight on mid-sized models with Phi-3.5-mini-instruct and similar 3-4 B-parameter models deliver ~88 % accuracy, offering strong performance at a fraction of the compute budget.

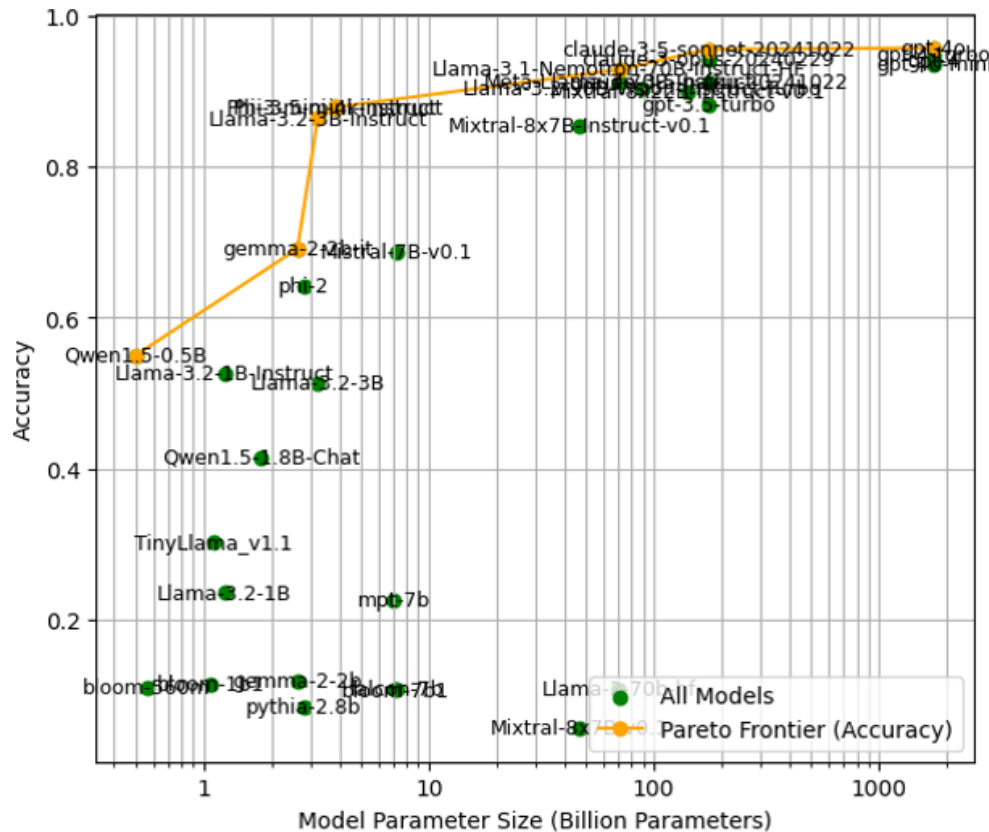
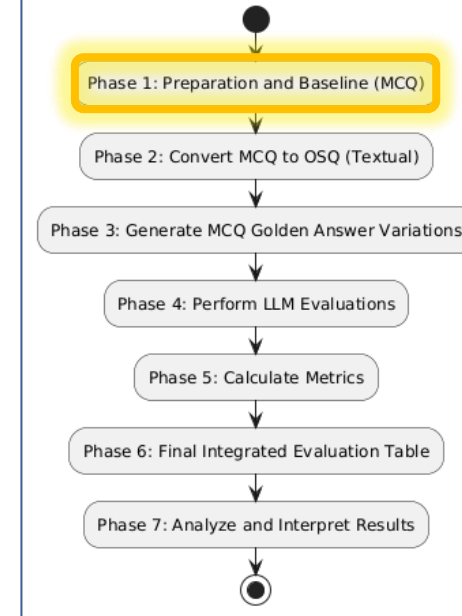


TABLE 4 Performance on SysEngBench by Category

LLM	Score ¹	SEO ²	LS ³	TP ⁴	TMPS ⁵	CCSEM ⁶	SEA ⁷
claude-3-5-sonnet-20241022	0.96	0.95	0.93	0.96	1.00	0.97	0.94
gpt-4o	0.96	0.95	0.95	0.96	0.97	0.96	0.97
gpt-4-turbo	0.95	0.93	0.92	0.95	1.00	0.97	0.95
gpt-4	0.94	0.92	0.92	0.95	0.97	0.96	0.97
gpt-4o-mini	0.94	0.91	0.93	0.94	0.99	0.96	0.98
claude-3-opus-20240229	0.94	0.92	0.92	0.95	0.99	0.95	0.97
Llama-3.1-Nemotron-70B-Instruct-HF	0.93	0.91	0.87	0.95	0.98	0.95	0.94
Llama-3.2-90B-Vision-Instruct-Turbo	0.91	0.89	0.86	0.92	0.99	0.89	0.92
Meta-Llama-3-70B-Instruct	0.91	0.90	0.87	0.93	0.98	0.88	0.96
claude-3-5-haiku-20241022	0.91	0.87	0.89	0.92	0.99	0.95	0.92
Meta-Llama-3.1-405B-Instruct-Turbo	0.90	0.91	0.84	0.91	0.91	0.88	0.89
Mixtral-8x22B-Instruct-v0.1	0.90	0.86	0.86	0.91	0.99	0.91	0.96
Phi-3-mini-4k-instruct	0.88	0.82	0.86	0.89	0.95	0.91	0.93
gpt-3.5-turbo	0.88	0.82	0.86	0.89	0.96	0.88	0.96
Phi-3.5-mini-instruct	0.88	0.81	0.85	0.89	0.97	0.92	0.93
Llama-3.2-3B-Instruct	0.87	0.82	0.82	0.88	0.96	0.85	0.95
Mixtral-8x7B-Instruct-v0.1	0.86	0.79	0.83	0.86	0.95	0.86	0.92
gemma-2-2b-it	0.69	0.59	0.76	0.68	0.83	0.71	0.73
Mistral-7B-v0.1	0.68	0.58	0.75	0.69	0.74	0.79	0.64
phi-2	0.64	0.54	0.81	0.61	0.66	0.67	0.69
Qwen1.5-0.5B	0.55	0.38	0.60	0.58	0.70	0.57	0.61
Llama-3.2-1B-Instruct	0.53	0.47	0.71	0.43	0.54	0.64	0.57
Llama-3.2-3B	0.52	0.43	0.76	0.44	0.58	0.66	0.43
Qwen1.5-1.8B-Chat	0.42	0.27	0.62	0.37	0.50	0.50	0.49
TinyLlama_v1.1	0.30	0.32	0.29	0.30	0.31	0.21	0.42
Llama-3.2-1B	0.24	0.28	0.27	0.19	0.30	0.26	0.19
mpt-7b	0.23	0.16	0.41	0.20	0.21	0.29	0.12
gemma-2-2b	0.12	0.14	0.19	0.09	0.06	0.20	0.06
Llama-2-70b-hf	0.11	0.13	0.17	0.09	0.05	0.19	0.03
bloom-1b1	0.11	0.12	0.19	0.10	0.04	0.20	0.03
bloom-560m	0.11	0.13	0.17	0.08	0.05	0.21	0.03
falcon-7b	0.11	0.13	0.17	0.09	0.05	0.19	0.03
bloom-7b1	0.11	0.12	0.19	0.08	0.06	0.20	0.03
pythia-2.8b	0.09	0.08	0.05	0.09	0.08	0.12	0.11
Mixtral-8x7B-v0.1	0.06	0.08	0.11	0.04	0.01	0.06	0.01

¹ Overall Score on SysEngBench
² Systems Engineering Overview
³ Lifecycle Stages
⁴ Technical Processes
⁵ Technical Management Processes
⁶ Cross-Cutting Systems Engineering Methods
⁷ Specialty Engineering Activities



Research Approach – Phase 2: Convert MCQ to OSQ

- MCQ questions will be filtered to see if they are candidates for conversion
- If the MCQ is a candidate, the question will be passed to a language model for conversion
- A filtering prompt and conversion prompt will be used for these steps

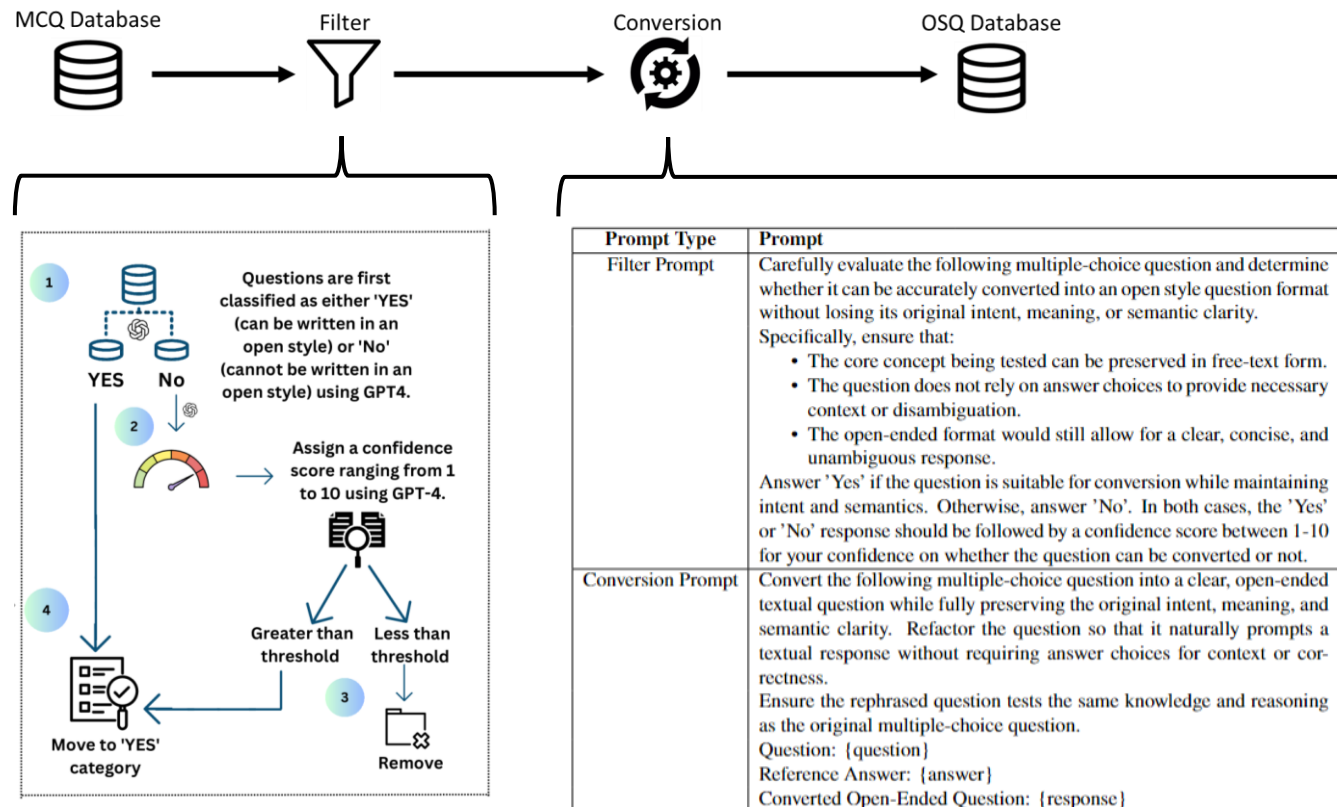
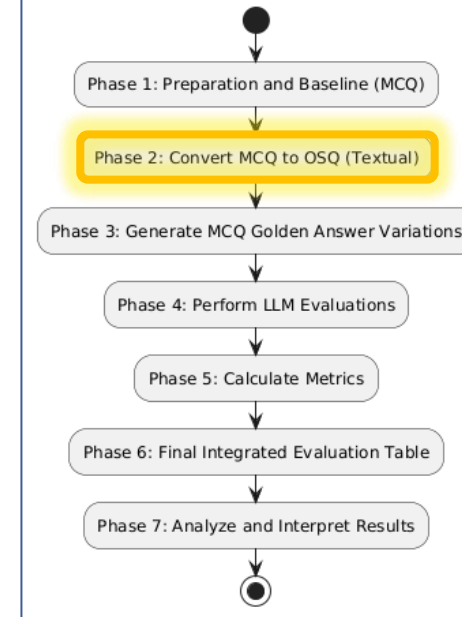
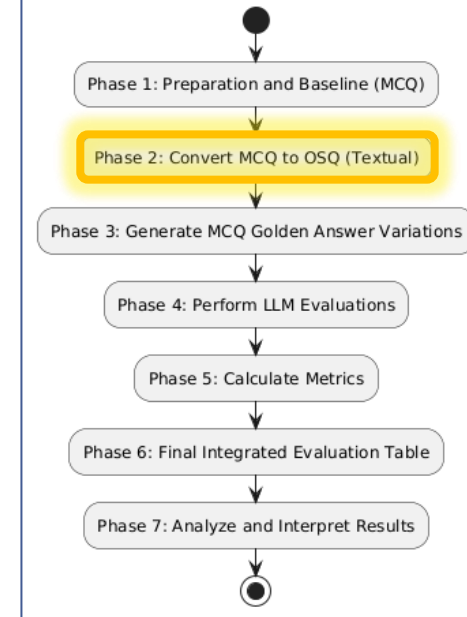
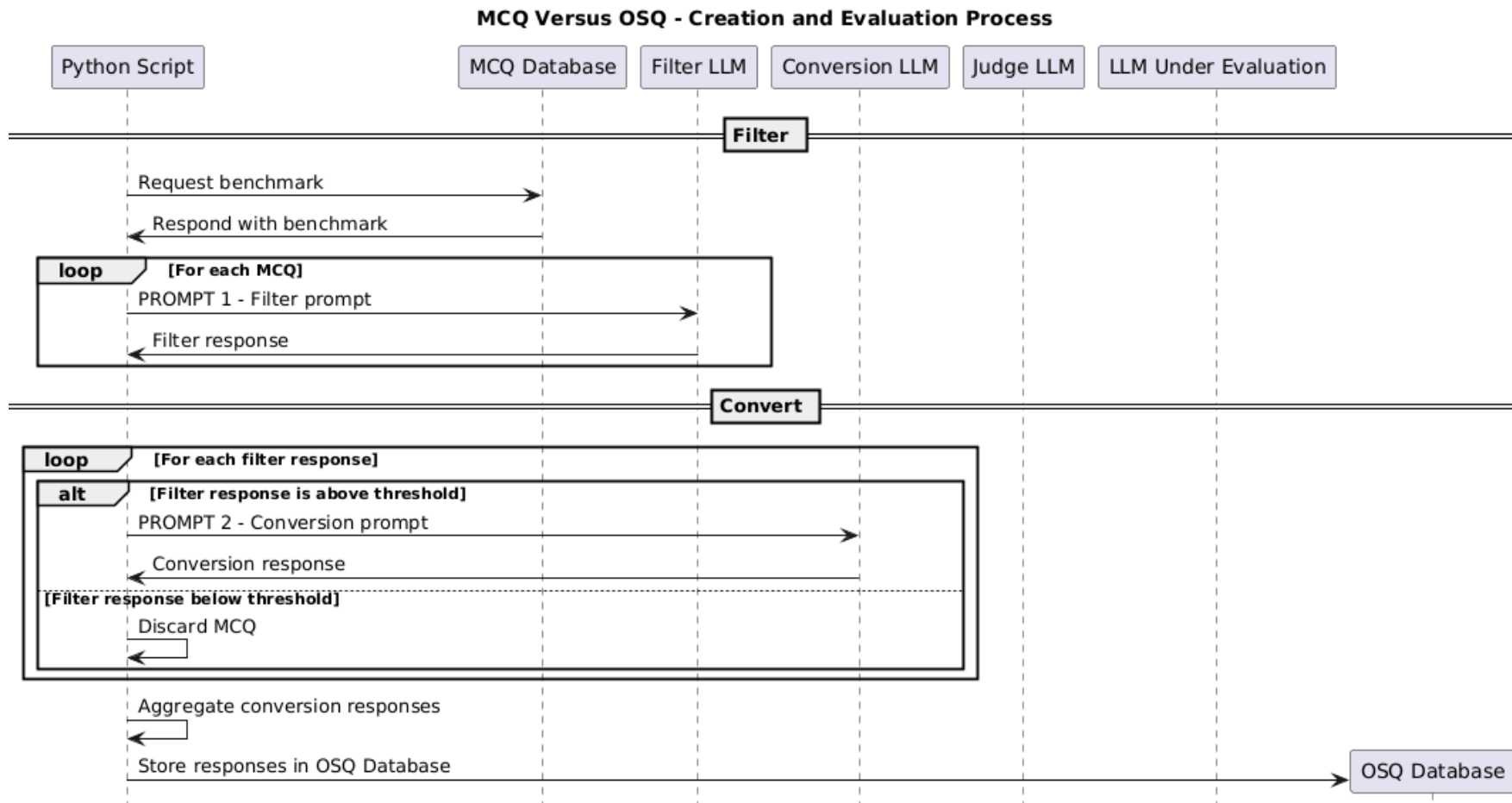


Table 2.2. Classification and Evaluation Prompts



Research Approach – Phase 2: Convert MCQ to OSQ

- Sequence diagram view for the conversion process

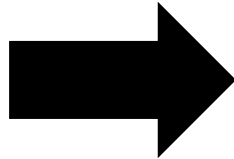


Research Approach – Phase 3: Generate MCQ Variations

- The existing, scattered golden answers will be aligned to different distractor variations

Scattered Golden Answers

	A	B	C	D
1.	☐	☐	☐	☒
2.	☐	☒	☐	☐
3.	☐	☐	☒	☐
4.	☒	☐	☐	☐



“A” Golden Answers

	A	B	C	D
1.	☒	☐	☐	☐
2.	☒	☐	☐	☐
3.	☒	☐	☐	☐
4.	☒	☐	☐	☐

“B” Golden Answers

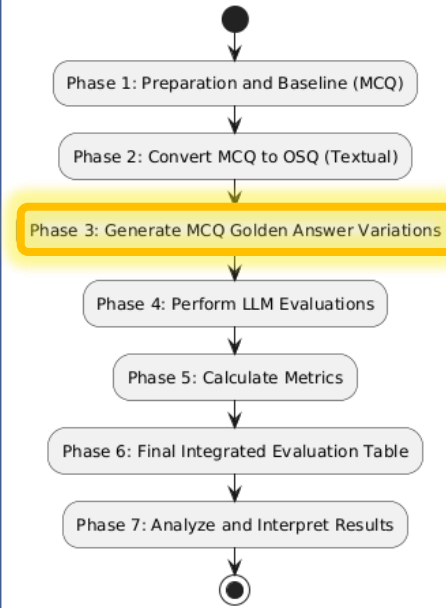
	A	B	C	D
1.	☐	☒	☐	☐
2.	☐	☒	☐	☐
3.	☐	☒	☐	☐
4.	☐	☒	☐	☐

“C” Golden Answers

	A	B	C	D
1.	☐	☐	☒	☐
2.	☐	☐	☒	☐
3.	☐	☐	☒	☐
4.	☐	☐	☒	☐

“D” Golden Answers

	A	B	C	D
1.	☐	☐	☐	☒
2.	☐	☐	☐	☒
3.	☐	☐	☐	☒
4.	☐	☐	☐	☒



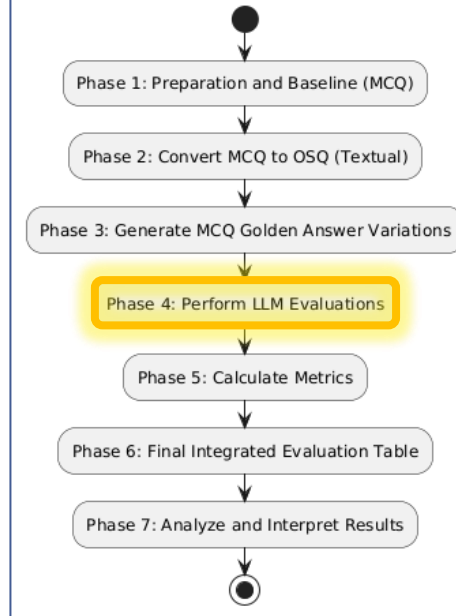
Research Approach – Phase 4: Perform Evaluations

- Both MCQ and OSQ tasks will leverage lm-eval-harness for systematic querying of the models
- Variations of the YAML file will be required for the MCQ distractor and OSQ variations

lm-evaluation-harness Script

```
!lm_eval \
  --model hf \
  --model_args pretrained={model_name} \
  --include_path ./ \
  --tasks sysengbench \
  --device cuda \
  --output output/sysengbench/ \
  --log_samples \
  --num_fewshot 0 \
  --batch_size auto \
  --gen_kwargs temperature=0.0
```

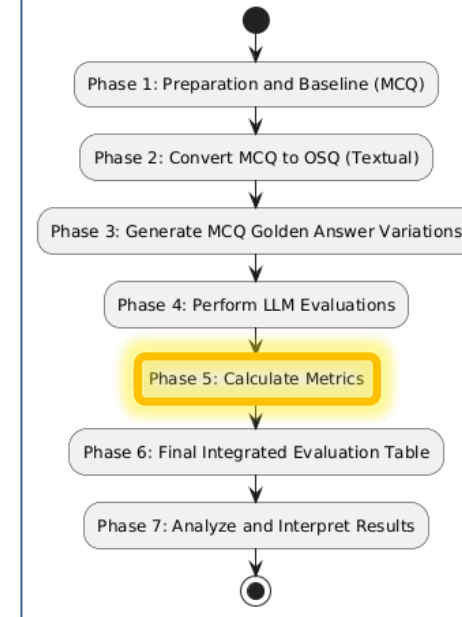
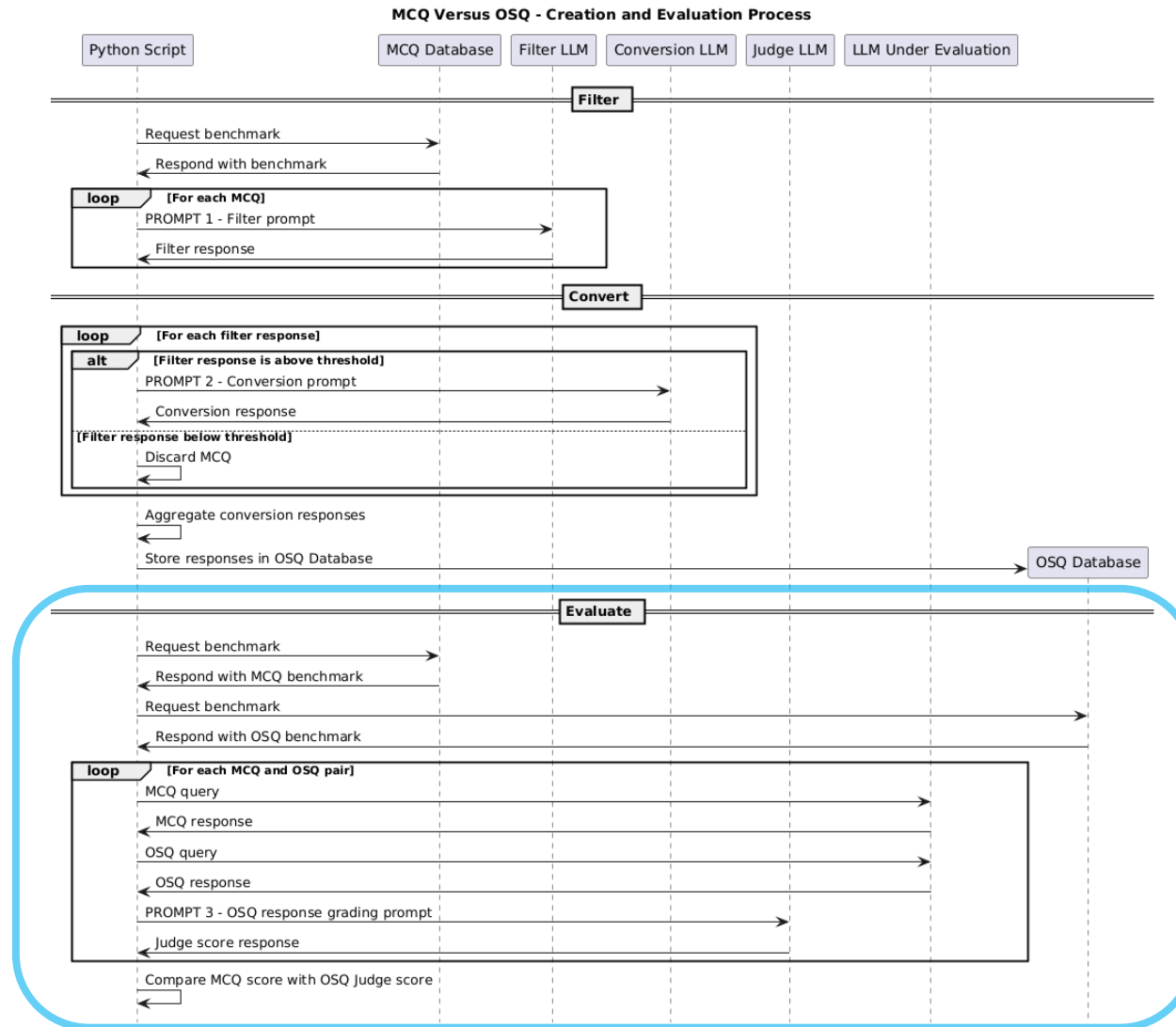
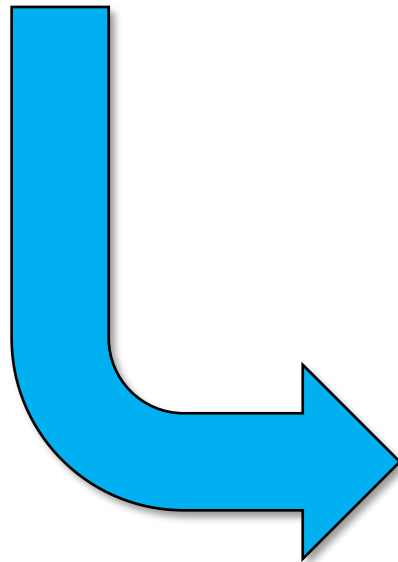
File Type	Loglikelihood Models (e.g. HF)	Textual Models (e.g. ChatGPT)
YAML File	<pre>task: sysengbench dataset_path: rabell/SysEngBench dataset_name: null output_type: multiple_choice training_split: null validation_split: null test_split: test doc_to_text: "Given the following question and four candidate answers (A, B, C and D), choose the best answer. Question: {{question}} A. {{choiceA}} B. {{choiceB}} C. {{choiceC}} D. {{choiceD}} Your response must consist solely of the single letter corresponding to the best answer, chosen from one of A, B, C or D. Any response other than a single letter (A, B, C, or D) will be considered invalid. Answer:" doc_to_target: "{{label}}" doc_to_choice: ["A", "B", "C", "D"] generation_kwargs: temperature: 0.0 metric_list: - metric: acc aggregation: mean higher_is_better: true - metric: acc_norm aggregation: mean higher_is_better: true metadata: version: 1.0 dataset_kwargs: trust_remote_code: true</pre>	<pre>task: sysengbench dataset_path: rabell/SysEngBench dataset_name: null output_type: generate_until training_split: null validation_split: null test_split: test doc_to_text: "Given the following question and four candidate answers (A, B, C and D), choose the best answer. Question: {{question}} A. {{choiceA}} B. {{choiceB}} C. {{choiceC}} D. {{choiceD}} Your response must consist solely of the single letter corresponding to the best answer, chosen from one of A, B, C or D. Any response other than a single letter (A, B, C, or D) will be considered invalid. Answer:" doc_to_target: "{{answer}}" generation_kwargs: temperature: 0.0 max_tokens: 20 until: - "</>" - "\n" filter_list: - name: "strict-match" filter: - function: "regex" regex_pattern: "([ABCD])" - function: "take_first" metric_list: - metric: exact_match aggregation: mean higher_is_better: true ignore_punctuation: true ignore_case: true metadata: version: 1.0 dataset_kwargs: trust_remote_code: true</pre>



Example Benchmark YAML Files

Research Approach – Phase 5: Calculate Metrics

- MCQ Grading
 - Simple accuracy check
- OSQ Grading
 - Using LLM as a Judge with different models (the variance for OSQ)



Research Approach – Phase 5: Calculate Metrics

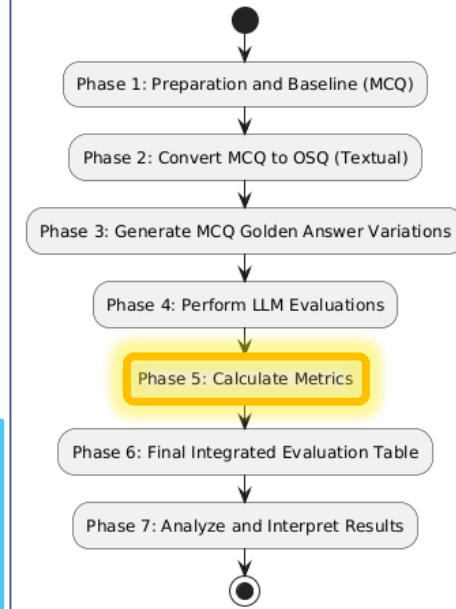
- MCQ Grading
 - Simple accuracy check
- OSQ Grading
 - Using LLM as a Judge with different models (the variance for OSQ)

Prompt Type	Prompt
Grading Prompt	You are an expert evaluator tasked with grading the model-generated response. Use the rubric (0-20 points per category, total 100): Rubric Categories: • Accuracy: Factually correct and supported? • Relevance: Fully addresses the question? • Clarity: Clear, logical, easy to follow? • Reasoning: Reasoning and insight? • Language: Well-written and precise? Return Format: Return your answer in the following format: Accuracy: {score}/20 – {justification} Relevance: {score}/20 – {justification} Clarity: {score}/20 – {justification} Reasoning: {score}/20 – {justification} Language: {score}/20 – {justification} Total Score: {sum}/100 Open-Ended Question: {question} Model Response: {response} Reference Answer: {reference}

Table 2.4. Grading Prompt with Rubric and Structured Return Format

Criterion	Score: 0-5 (Poor)	Score: 6-10 (Needs Improvement)	Score: 11-15 (Good)	Score: 16-20 (Excellent)
Accuracy and Factual Correctness	Major inaccuracies or fabrications dominate the response.	Some correct information, but several significant errors or unsupported claims are present.	Mostly accurate with minor factual issues or missing details.	Completely accurate, factually correct, and well-supported by knowledge.
Relevance and Completeness	Off-topic or fails to address the question; response is incomplete.	Some relevant content, but important aspects are missing or underdeveloped.	Mostly relevant and covers key elements with acceptable depth.	Fully relevant and comprehensive, addressing all parts of the question thoroughly.
Clarity, Coherence, and Organization	Poorly structured, incoherent, and difficult to follow.	Somewhat clear but with disorganized sections or confusing flow.	Mostly clear, logical, and easy to follow with minor structural issues.	Exceptionally clear, coherent, and well-organized throughout.
Depth of Reasoning and Insight	Lacks reasoning, analysis, or logical argumentation.	Basic reasoning present but shallow, with weak or unsupported analysis.	Solid reasoning and some analytical depth with supporting examples.	Strong, well-supported reasoning and insightful analysis throughout.
Language Quality, Grammar, and Precision	Frequent grammar errors, poor phrasing, verbose or imprecise writing.	Understandable but with noticeable grammar or language issues.	Generally well-written, concise, with minor language flaws.	Grammatically flawless, precise, professional, and stylistically strong.

Table 2.3. Notional Rubric for LLM as a Judge in Open-Style Question Evaluation

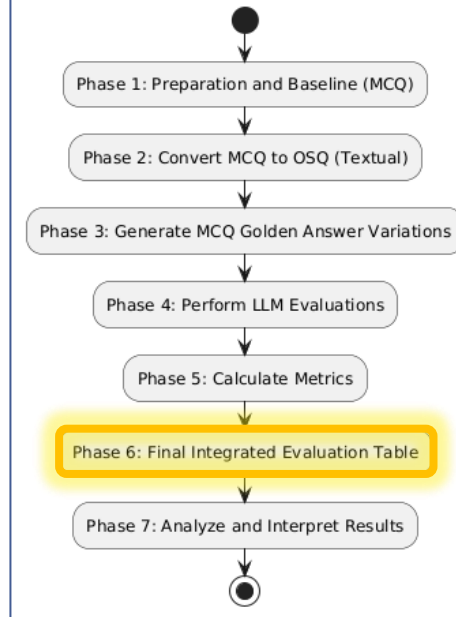


Research Approach – Phase 6: Aggregate Data to Tables

- Aggregate data from both MCQ and OSQ tasks
- Calculate a multitude of metrics: accuracy, variance, intraclass correlation coefficient (ICC) if multiple LLMs-as-a-Judge
- Calculate statistical test metrics as applicable

SysEngBench Evaluation Table

SysEngBench Category	MCQ (Accuracy by Golden Answer)				Textual Evaluation (Score by LLM Graders)			Metrics	
	A	B	C	D	LLM-A	LLM-B	LLM-C	MCQ Accuracy Variance	Textual Mean (ICC)
Stakeholder Needs	0.80	0.65	0.55	0.75	88	84	91	0.0119	87.7 (0.83)
Requirements Engineering	0.90	0.88	0.85	0.92	93	89	94	0.0011	92.0 (0.91)
Architecture Design	0.60	0.40	0.35	0.55	79	73	76	0.0138	76.0 (0.78)
Implementation	0.85	0.60	0.50	0.80	84	80	82	0.0229	82.0 (0.85)
Integration	0.75	0.45	0.40	0.70	77	72	75	0.0238	74.7 (0.76)
Verification & Validation	NOTIONAL OUTPUT USING SYNTHETIC DATA FOR PROTOTYPING PURPOSES								80.0 (0.84)
Transition									84.7 (0.87)





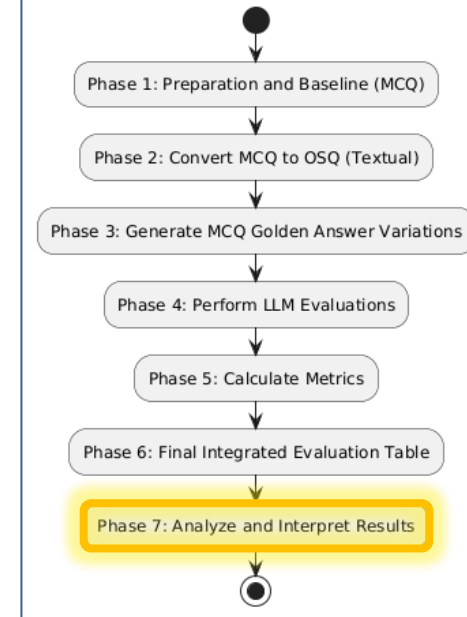
Research Approach – Phase 7: Analysis Methods

Several facets of the data will be explored, including but not limited to:

- How does MCQ fair against OSQ for accuracy? For variance?
- What topics do models perform better at MCQ than OSQ? OSQ than MCQ?
- Which models have high correlation between doing well at MCQ and OSQ?
- Is it clear what topics/concepts in SE need what evaluation modality?

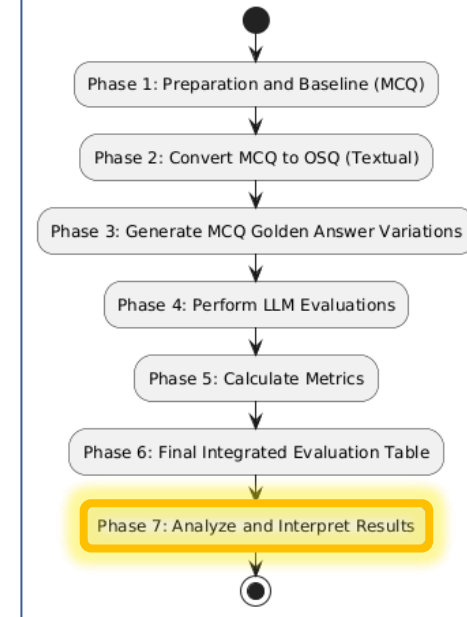
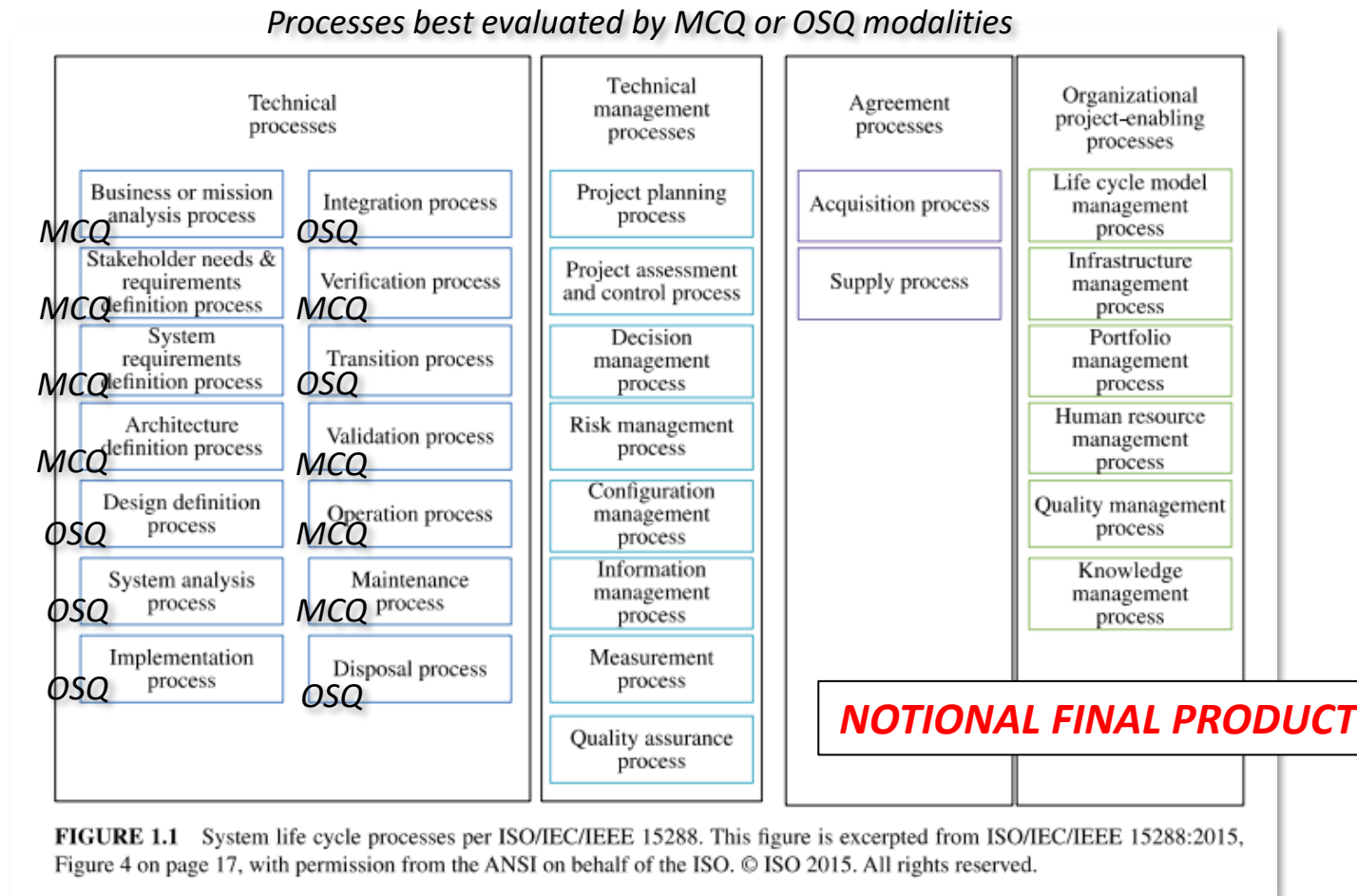
When it comes to interpreting the summary table:

- Categories with high variance indicate guessing behavior in MCQ, suggesting OSQ/textual or richer methods may be superior.
- Categories with low variance and high MCQ accuracy might reliably use MCQ evaluation.
- Categories with high ICC scores show textual evaluations are consistent and valuable; lower ICC indicates complexity or grader disagreements.



Final Product: Practical recommendations and insights on the best evaluation format per Systems Engineering task category.

Research Approach – Phase 7: Analysis Methods



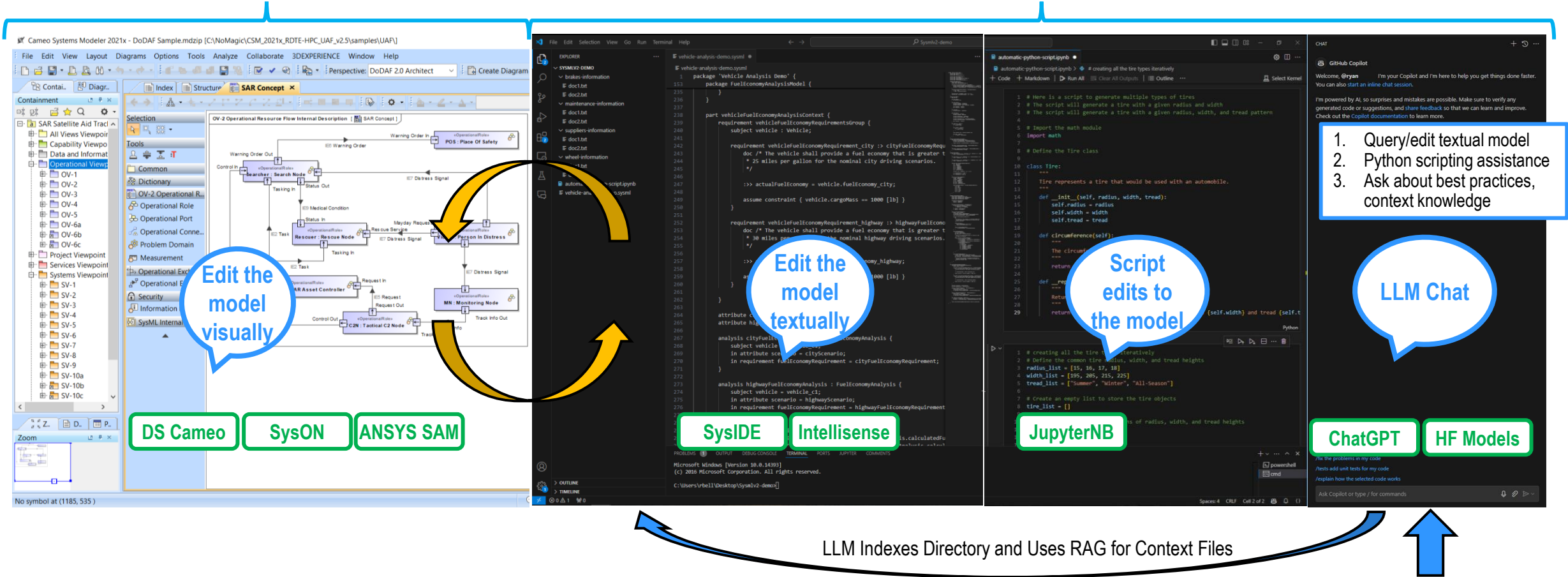


Research Contributions

- **Task-Modality Alignment Framework for Systems Engineering Evaluation**
 - *Maps SE task types to the most appropriate evaluation format (MCQ vs OSQ) and advocates hybrid benchmarks to capture both objectivity and deep reasoning.*
- **Robustness Analysis of MCQ Evaluation Using Distractor Variation for Systems Engineering**
 - *Systematically rotates the correct answer among distractors; accuracy variance becomes a diagnostic for whether models truly understand concepts or are merely guessing.*
- **Extension of SysEngBench with Multi-Format and Parallel Evaluation Items**
 - *Adds matched MCQ + OSQ questions via an LLM-in-the-loop pipeline, enabling controlled “apples-to-apples” comparisons on identical SE knowledge.*
- **Comparative Effectiveness of MCQ, OSQ, and Hybrid Evaluation Modalities**
 - *Evaluates strengths and weaknesses of each format across SE task types and proposes hybrid strategies (e.g., distractor-variant MCQ plus consensus-scored OSQ) for higher-fidelity assessment.*
- **Evidence-Based Guidance for Practical LLM Integration in Systems Engineering Workflows**
 - *Translates evaluation insights into actionable recommendations, showing which modalities best predict reliable model performance for real-world SE tasks.*

Package Tree and Visual Diagramming
(Cameo, other vendors, or Open Source)

Directory, SysMLv2 Textual, Python Scripting, Command Line, and Language Model Chat
(VSCode, Python, Jupyter, and Continue Extensions, Open Source)



- Backend is Cloud/Local Inference Server w/ Vector Database
- Cloud Model = COTS approved / Hosted Fine-tuned
- Local Model = Fine-tuned from this research
- Model = Single or AI Team!

Execution Plan: Dissertation Outline

Section ID	Chapter Title	Description
I - VII	Front Matter	- Abstract - Executive summary
1	Introduction	- Background & Motivation - Identify the Problem - Research Questions and Objectives - Research Methodology - Research Contributions, and Impact - Scope, Boundaries, and Limitations - Structure of Dissertation
2	Literature Review	- Introduction and Context Setting - Current Benchmarking Landscape - Evaluation Methods for LLMs - Unique Evaluation Challenges in SE and Cross-Domain Strategies - Experimental Design for Benchmark Comparison - Measurement and Analysis Techniques.
3	Materials and Methods	- Research Framework - Dataset Construction - Evaluation Methods and Computations - Analysis Metrics and Methods
4	Results	- Performance on Multiple Choice Questions - Performance on Open Style Questions - Comparison Across Modalities - Mapping to INCOSE Handbook Categories
5	Discussion	- Interpretation of Findings - Implications for Benchmarking Methodologies - Practical Applications for Systems Engineering and Insights for LLM Employment - Limitations and Future Work
6	Conclusion	- Summary of Contributions - Final Thoughts

Table 4.1. Proposed Dissertation Outline

Task ID	Description	Completion Date (Month, Year)
T.1	Develop Dissertation Outline	March 2025
T.2	Complete Dissertation Proposal	May 2025
T.3	Draft Research Questions and Objectives (Chapter 1)	May 2025
T.4	Submit Journal 1	March 2025
T.5	Complete Literature Review (Chapter 2)	July 2025
T.6	Complete Materials and Methods (Chapter 3)	August 2025
T.7	Submit Journal 2	March 2025
T.8	Conduct Experiments to get Results	September 2025
T.9	Complete Results and Discussion (Chapter 4 and 5)	October 2025
T.10	Complete Conclusion (Chapter 6)	October 2025
T.11	Complete Dissertation Formatting and Submit Initial Draft	October 2025
T.12	Defend Dissertation	November 2025
T.13	Complete Final Edits and Submit to TPO and Department Chair	December 2025
T.14	Route Final Draft	December 2025
T.15	Graduate	March 2026

Table 4.2. Proposed Dissertation Schedule



Execution Plan: Resources and Availability Assessment

- The resources required to execute the dissertation work are currently available
- An assessment of resource availability is conducted monthly to address the impact on the executed work
- Any emergent resource needs are then identified, an impact analysis to the dissertation, and course of action are reported to the dissertation advisor and committee for concurrence
- The sponsoring command provides funding for tuition and travel to campus, which supports the research activities for concurrence

Resources are available to perform the work!



Execution Plan: Notional Publications

Working Title	Publication Type and Venue	Date	Author Type
Baselining AI's Systems Engineering Performance in Cost Modeling	Presentation: Boehm CSSE Annual Research Review 2024	Apr 2024	Primary
Introducing SysEngBench: A Novel Benchmark for Assessing Large Language Models in Systems Engineering	Paper & Panel: NPS ARS 2024	May 2024	Primary
Leveraging Generative AI to Modify and Query MBSE Models	Paper & Panel: NPS ARS 2024	May 2024	Secondary
Using AI Tools for SE, from Requirements Generation and Management to Risk Identification, Analysis, and Management	Presentation: INCOSE IS 2024	Jul 2024	Primary
Tailored Learning for Cost Modeling: An Open-Source RAG-Based Tool	Presentation: 2024 BCSSE COCOMO Forum	Sep 2024	Primary
(In Review) Baselining Large Language Model Performance in Systems Engineering Using SysEngBench	Paper: Wiley INCOSE Journal AI in SE Special Issue	Feb 2025	Primary
(In Review) Balancing Accuracy and Cost: Trade-Offs in Large Language Model Quantization for the Systems Engineering Domain	Paper: Wiley INCOSE Journal AI in SE Special Issue	Feb 2025	Primary
PrivateAIDELPHI: Adopting and Adapting Private AI for Risk Assessment of Safety-Critical Systems	Paper & Poster: RAMS 2025	Feb 2025	Secondary
A Generative AI-Driven Systems Engineering Maturity and Cost Modeling Framework	Paper: CSER	Apr 2025	Secondary
Exploring Visual Question-Answer Capabilities of Multi-Modal Large Language Models with Model-Based Systems Engineering Models	Paper: NPS ARS 2025	May 2025	Secondary
Automating AI Expert Consensus: Feasibility of Language Model-Assisted Consensus Methods for Systems Engineering	Paper: NPS ARS 2025	May 2025	Primary
(In Work) Cognitive Probing of Multi-Modal LLMs: A Comparative Study of Question Formats on SysML v1 Structural Diagrams	Paper: Wiley INCOSE Journal	Jun 2025	Secondary
(Accepted) The Cost of Expertise: Performance Trade-Offs in Fine-Tuned LLMs for Systems Engineering	Paper: INCOSE IS 2025	Jul 2025	Secondary
(In Work) Evaluating Structured Model Output in LLMs: A Case Study in SysML v2	Paper: Wiley INCOSE Journal	Sep 2025	Secondary



Committee Management Plan: The Committee

- **Essential Dissertation Supervisor, Chair, and PhD Candidate Advising Agreements**

Dr. Madachy

- **We interact early and often.** We reserve a bi-weekly or weekly meeting time on our calendars. A meeting may be cancelled and replaced by an email update, but every other week, there will be either a meeting or an emailed status.

- **Essential Dissertation Committee and PhD Candidate Advising Agreements**

Dr. Giachetti, Green, Kolsch, Van Bossuyt

- **We interact routinely.** We reserve a quarterly meeting time on our calendars.
- **We respect each other's time.** Meetings are held with entrance and exit criteria to encourage progress. If we are unable to make a meeting or deliverable commitment, we inform as soon as we are aware.
- **We recognize that the journey is a process, not an event.** We don't forget or ignore fundamental needs; we tend to these and return refreshed.
- **We view mistakes with interest, not deprecation.** We learn far more from our own imperfect models than from studying someone else's perfect model.
- **We choose to be here.** We give our all to advance and promote better system design, not just because we get another professional credential, but because we love solving interesting problems that advance the practices of systems engineering and design.

- **Your Dissertation Committee's Commitments to You**

- To have strong interest in your proposed research topic upon agreeing to advise.
- To be present and available for regular meetings.
- To engage students to help each other, where relevant and practical, based on the premise that mastery of a subject is enhanced and reinforced in explaining it to others.
- To help you scope your research given the target timelines, resources, and graduation dates.
- To take on only the number of students that allows for sufficient attention to each.



Committee Management Plan: The Student

■ Your Commitments to Your Dissertation Committee

- To love the topic you choose to take on. This passion will be your motivator when the going gets tough, and it will at times!
- To give your research high priority, and make every effort to follow the schedule agreed to by your dissertation supervisor, chair, and committee. One-to-many advising works when the advising is slotted into blocks of time with start and end dates. If you exceed the agreed timeline, this has an impact on workload plans and on other students seeking advisors. You must be prepared for the possibility that you may need to change your topic and/or find a new dissertation supervisor, chair, or committee if you do not complete your dissertation within the agreed timeline.
- To support or accept the support of other students in the spirit of collaboration. Teaching others is the best way to deepen your knowledge and experience with a subject; students learn on both sides of the interaction.
- To relentlessly ask questions until you understand the concepts, seeking answers first through independent search, then through reaching out to others.



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Questions?

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